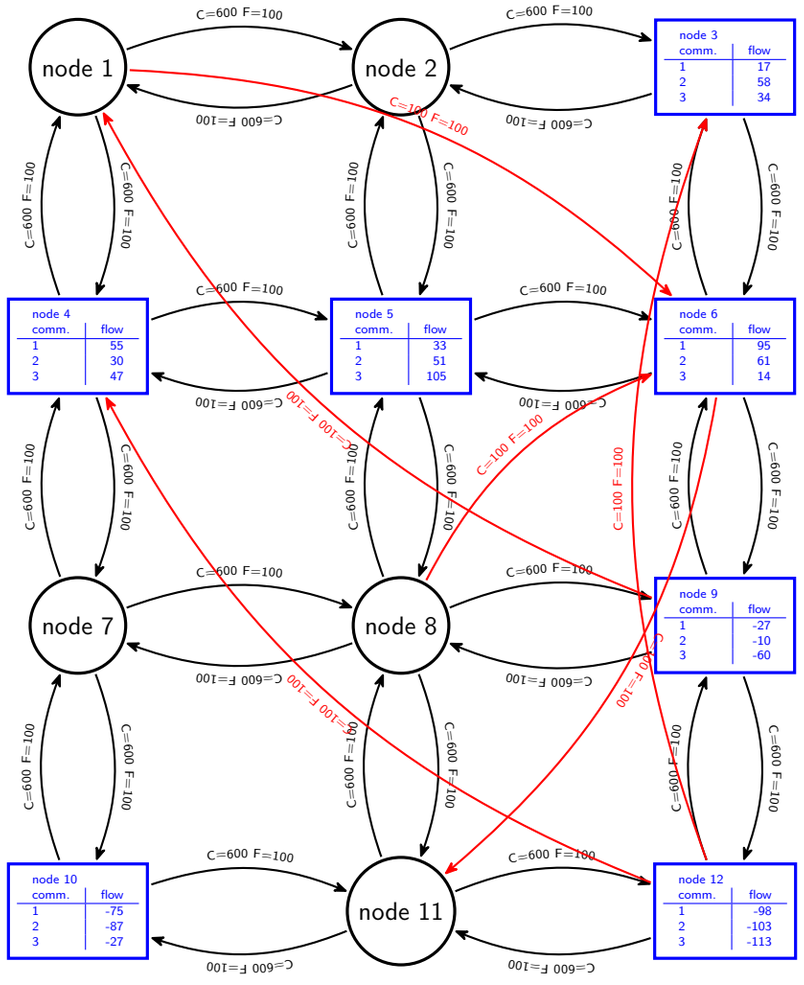
# 1 Generated network based on grid topology



## 2 Notes

- Black arcs are part of the base, grid secondary topology.
- Red arcs are additional, randomly generated.
- $C = x$: overall capacity of arc is equal to $x$.
- $F = y$: fixed cost of arc is equal to $y$.
- See below for capacities and variable costs per arc, per commodity.
- Circular nodes are transit nodes.
- Rectangular nodes are source ($flow > 0$) or sink ($flow < 0$) nodes for stated commodities.

## 3 Capacities and variable costs per arc, per commodity

| Arc | Comm. | Cap. | Var. cost | Comm. | Cap. | Var. cost | Comm. | Cap. | Var. cost |
|---|---|---|---|---|---|---|---|---|---|
| (1, 2) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (1, 4) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (1, 6) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (2, 1) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (2, 3) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (2, 5) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (3, 2) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (3, 6) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (4, 1) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (4, 5) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (4, 7) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (5, 2) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (5, 4) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (5, 6) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (5, 8) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (6, 3) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (6, 5) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (6, 9) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (6, 11) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (7, 4) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (7, 8) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (7, 10) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (8, 5) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (8, 6) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (8, 7) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (8, 9) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (8, 11) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (9, 1) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (9, 6) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (9, 8) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (9, 12) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (10, 7) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (10, 11) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (11, 8) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (11, 10) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (11, 12) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (12, 3) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (12, 4) | 1 | 100 | 10 | 2 | 100 | 10 | 3 | 100 | 10 |
| (12, 9) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |
| (12, 11) | 1 | 200 | 10 | 2 | 200 | 10 | 3 | 200 | 10 |